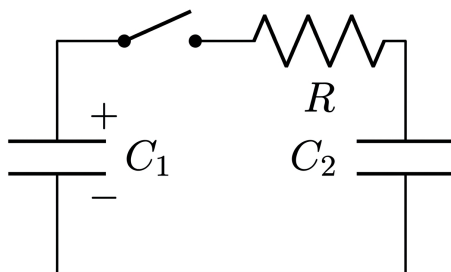


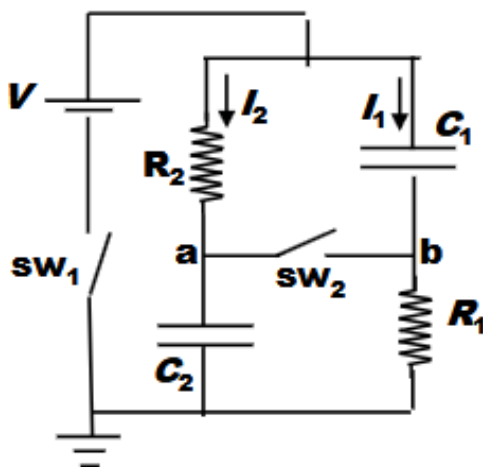
Week 11 Discussion Problems

Topics: Circuits, Magnetic Force

- Consider the circuit shown below that contains capacitors C_1 and C_2 and resistor R . Initially the switch is open and the capacitor C_1 has charge Q_0 . The switch is closed at $t = 0$.
 - What is the initial current that flows through the resistor right after the switch is closed ?
 - After a very long time, what are the charges Q_1 and Q_2 on C_1 and C_2 respectively ?
 - What is the charge $Q_2(t)$ on C_2 as a function of time ?
 - What is the current flowing into C_2 as a function of time ?
 - How will $Q_2(t)$ be modified if a dielectric of constant ϵ is inserted in between the plates of capacitor C_2 ?



- The known variables are V , R_1 , R_2 , C_1 , and C_2 . Consider the following circuit.

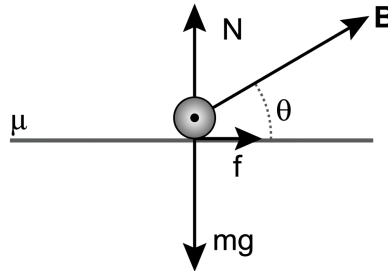


Before $t = 0$, both switches are open and the capacitors are uncharged. At time $t = 0$, we close switch 1. Switch 2 remains open.

- At time $t = 0$, what are the currents I_1 and I_2 in each arm?
- At time $t = 0$, what is the potential difference between point a and point b?
- At time $t = \infty$, what is the potential difference between point a and point b?

Now, consider the circuit in the state as it would be in part c. Next, we close switch 2, so that now both switches are closed.

- (d) What is the final potential of point b with respect to ground at $t = \infty$?
- (e) How much does the charge change on each capacitor at $t = \infty$ (compared to just before switch 2 was closed) ?
3. A copper rod of mass m rests on two horizontal rails a distance d apart, and carries a current i_0 from one rail to the other. The coefficient of static friction is μ . A uniform magnetic field of unknown magnitude B points upward at angle θ . Here is a side view:



In this diagram, the current i_0 in the wire is coming out of the page, and $\mathbf{f}$ denotes the friction force. Notice that the magnetic field vector $\mathbf{B}$ is *not* a force vector.

- (a) The magnetic force on the wire has not been included in the above diagram. What is the magnitude of this force? Sketch the force vector on the diagram, and break it into horizontal and vertical components.
- (b) If the wire is just on the verge of sliding sideways, then how large is the magnetic field B ? Your answer should involve θ .
- (c) By minimizing your expression for B with respect to θ , find the smallest magnetic field that could cause the wire to slide sideways.
- (d) If the magnetic field were *vertical*, which direction would the force on the wire point? Why is the optimal magnetic field found in part (c) not vertical ?
4. Find the net magnetic force on the whole wire of current I which consists of two straight segments of length L and a half circular segment of radius R as shown below. The magnetic field is constant and in the $\hat{z}$ direction.

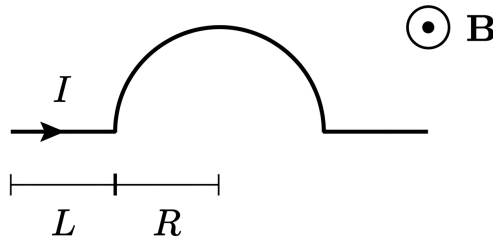


Figure 1: Wire with two straight segments and one semi-circular segment.